

Saifeldin Yousry Ahmed

Computer Engineering Student | AI/ML Engineer

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Objective

Junior AI/ML Engineer with hands-on experience in developing and deploying intelligent systems. Skilled in machine learning, deep learning, NLP, and large language models with a proven track record of delivering scalable AI solutions. Passionate about advancing AI-driven innovations that combine data science, ML engineering, and software systems to solve real-world problems and drive business impact.

Education

Ain Shams University International Program, BSc in Computer Engineering and Software Systems Oct 2021 – July 2026

- Dual Degree Program with **University of East London** (UEL)
- Specializing in Data Science and AI

Dr. Nermein Ismail International Schools, International Baccalaureate (IB) Diploma Graduated 2021

- Valedictorian

Experience

Xyris Digital Solutions, Junior AI Engineer Sept 2025 – Present

- Engineered enterprise RAG and agentic systems using LangChain, LangGraph, and Hugging Face Transformers, integrating FAISS and ChromaDB for scalable semantic retrieval and dynamic context routing across large knowledge bases.
- Built conversational AI and workflow agents with FastAPI, LangGraph state machines, and IBM watsonx.ai, including real-time interview transcription and evaluation pipelines using STT models (Whisper) and LLM scoring agents for structured assessments.
- Designed automation and orchestration pipelines using Sentence Transformers for embedding generation, ChromaDB for retrieval, and SQLite/stateful graphs for stepwise logic, supporting reliable multi-stage processing across internal AI workflows.
- Developed internal tools and NLP processing interfaces with Streamlit and GPT models for semantic analysis, classification, extraction, and content restructuring; regularly participated in high-end client meetings to gather requirements, present architectures, and align AI solutions with enterprise needs.

Xyris Digital Solutions El Sewedy Electric, AI/ML Intern July 2025 – Sept 2025

- Worked on fine-tuning Transformer-based models using Hugging Face and PyTorch, covering dataset preparation, domain adaptation, and evaluation for NLP tasks including classification, extraction, and summarization.
- Developed ML pipelines for predictive maintenance using enterprise-grade machinery and SCADA sensor data handling preprocessing, feature engineering, validation loops, and benchmarking to support model reliability and production readiness.

Huawei AI Training Program, Trainee Jan 2025 – March 2025

- Acquired foundational expertise in Python, ML/DL algorithms, and Huawei AI frameworks through hands-on projects, focusing on model design and deployment to solve real-world optimization challenges.

Alstom Egypt for Transport Projects, Intern July 2024 – Sept 2024

- Rotated through key engineering departments, contributing to system integrations that strengthened safety protocols and mitigated risks by identifying and addressing vulnerabilities in control mechanisms.

Skills

Programming & Scripting: Python, SQL, C/C++ , Java, JavaScript, HTML, CSS, R

AI/ML Frameworks: TensorFlow, Keras, Pytorch, Scikit-Learn, LangChain, **LangGraph**, Hugging Face Transformers, OpenAI API, IBM Watsonx.ai, FastAPI, Streamlit, Sentence Transformers

Data Science & Analysis: Pandas, NumPy, Matplotlib, Seaborn, Data Preprocessing, Feature Engineering, RAG Systems, Prompt Engineering, Vector Databases (e.g., FAISS, ChromaDB), Databases (SQLite)

Personal Skills: Highly Adaptive, Problem Solver, Strong Communicator, Effective Leader, Continuous Learner

Languages: English (Full Proficiency, *IELTS Certified*), Arabic (Native)

Certificates

HCIA-Artificial Intelligence V3.5

Issued by: Huawei Talent and NTI

Projects

Real-Time Face Analysis (Age, Gender, Emotion)

Used CNNs and MobileNetV2 to detect emotions (FER2013), age, and gender (UTKFace) from webcam streams in real time. Achieved smooth, responsive prediction UI with live annotations.

Technologies: TensorFlow, Keras, OpenCV, Python

github.com/SaiffMoh/RealTime-Face-Analysis-Age-Gender-Emotion

Twitter Sentiment Analysis

Collected live tweets via Tweepy and analyzed sentiment using Scikit-Learn pipelines. Visualized public opinion and performed emotion classification.

Technologies: Python, Tweepy, Scikit-Learn, Pandas

github.com/SaiffMoh/Twitter-Sentiment-Analysis

BarCodeVision – Barcode Defect Detection & Repair

Designed a computer vision pipeline to detect and correct damaged barcode images using OpenCV preprocessing before barcode extraction.

Technologies: Python, OpenCV, NumPy

github.com/SaiffMoh/BarCodeVision

Distributed Web Crawler (CSE354)

Designed a fault-tolerant distributed web crawler that fetches and indexes pages concurrently using socket communication. Implemented with peers in Python.

Technologies: Python, Sockets, Threading, BFS Crawling

github.com/Saif2310/CSE354-Distributed-Web-Crawler

DHCP Network Simulator (CSE381)

Simulated DHCP protocol behavior in a computer network environment. Implemented IP address allocation, lease renewal, and client-server negotiation.

Technologies: C++ , Network Protocols

github.com/SaiffMoh/DHCP-Computer-Network-Project